

temp ht above humidity % Gr. cover
 Sea level

Delhi

Mumbai

Chennai

Kolkata

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} + \begin{bmatrix} 0 & 0 & 1 \\ 2 & -1 & \sqrt{2} \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 1+0 & 2+0 & 3+1 \\ 4+2 & 5-1 & 6+\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 4 \\ 6 & 4 & 6+\sqrt{2} \end{bmatrix}$$

$$(A+B)(i,j) = A(i,j) + B(i,j)$$

$$\begin{aligned} i &= 1 \dots m \\ j &= 1 \dots n \end{aligned}$$

$$= B(i,j) + A(i,j)$$

$$= (B+A)(i,j)$$

Comm.

$$(A+B)+C = A+(B+C) \rightarrow \text{asso}$$

$$A \rightarrow m \times n$$

$$B = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & & \\ 0 & \dots & 0 \end{bmatrix}_{m \times n} \rightarrow \text{Zero matrix of order } m \times n$$

$$A + 0 = A$$

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

$$\Rightarrow A + (-A) = 0$$

$$-A = \begin{bmatrix} -a_{11} & \cdots & -a_{1n} \\ \vdots & & \vdots \\ -a_{m1} & \cdots & -a_{mn} \end{bmatrix}$$

↓
additive
inv

Scalar multiplication

$$k \cdot A = \begin{bmatrix} ka_{11} & ka_{12} & \cdots \\ \vdots & \vdots & \ddots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

$$2 \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 0 \end{bmatrix}$$

$$1 \cdot A = A$$

$$k \cdot (A + B) = k \cdot A + k \cdot B$$

$$(k + l) \cdot A = k \cdot A + l \cdot A$$

(

$$\begin{array}{c} A \quad B \\ m \times n \quad p \times n \\ \downarrow \quad \downarrow \\ L = \downarrow \\ m \times n \end{array}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$2 \times 3 \quad \quad \quad 2 \times 2$
L $\neq$ J

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$2 \times 2 \quad \quad \quad 2 \times 3$

$$= \begin{bmatrix} \underline{1} & \underline{2} & \underline{3} \\ \underline{6} & 9 & 12 \end{bmatrix}_{2 \times 3}$$

$$A: m \times \underline{n} \quad B: \underline{n} \times p$$

$$(AB)_{(i,j)} = \sum_{k=1}^n A(i,k) B(k,j)$$

$$\left\{ \begin{matrix} i=1, \dots, m \\ j=1, \dots, p \end{matrix} \right.$$

$$\begin{matrix} c_1 & c_2 & c_3 \\ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$= \begin{bmatrix} b_1 + 2b_2 + 3b_3 \\ 4b_1 + 5b_2 + 6b_3 \end{bmatrix}$$

$$= \begin{bmatrix} b_1 \\ 4b_1 \end{bmatrix} + \begin{bmatrix} 2b_2 \\ 5b_2 \end{bmatrix} + \begin{bmatrix} 3b_3 \\ 6b_3 \end{bmatrix}$$

$$= b_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + b_2 \begin{bmatrix} 2 \\ 5 \end{bmatrix} + b_3 \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$= b_1 c_1 + b_2 c_2 + b_3 c_3$$

$$A = [A_1 \ A_2 \ \dots \ A_n]$$

$$A \begin{matrix} \times \\ m \times n \quad n \times 1 \end{matrix}$$

$$= x_1 A_1 + x_2 A_2 + \dots + x_n A_n$$

$$\begin{matrix} A & B \\ \downarrow & \downarrow \\ m \times n & n \times m \end{matrix} \quad \begin{matrix} B & A \\ \downarrow & \downarrow \\ n \times m & m \times n \end{matrix}$$

$$n = m$$

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0$$

$$AB = 0 \not\Rightarrow A = 0 \text{ or } B = 0$$

$$(A+B)^T = A^T + B^T$$

$$\begin{aligned} & \underline{(A+B)^T(i,j)} \\ &= (A+B)(j,i) \\ &= A(j,i) + B(j,i) \\ &= A^T(i,j) + B^T(i,j) \\ &= (A^T+B^T)(i,j) \end{aligned}$$

$$\begin{aligned} & (A^T)^T(i,j) \\ &= A^T(j,i) \\ &= A(i,j) \end{aligned}$$

$$A^{TT} = A$$

$$(AB)^T = B^T A^T$$

$$\begin{aligned} & \quad \quad \quad \hookrightarrow \text{Try proving} \\ \overline{a+ib} &= a-ib \quad i=\sqrt{-1} \text{ this!} \end{aligned}$$

$$A^H = \bar{A}^T$$

$$\underline{\underline{[1 \quad 1-i]^H = [1 \quad 2-i]}}$$

$$p = \underbrace{\begin{bmatrix} 1 & 1-i \\ 2+i & 1+3i \end{bmatrix}}_A^H = \begin{bmatrix} 1 & 2-i \\ 1+i & 1-3i \end{bmatrix}$$

$$\text{sym} \Rightarrow A = A^T$$

$$\bar{A} = \begin{bmatrix} 1 & 1+i \\ 2-i & 1-3i \end{bmatrix}$$

$$\bar{A}^T = \begin{bmatrix} 1 & 2-i \\ 1+i & 1-3i \end{bmatrix} = A^H$$

$$A^H = A \rightarrow A \text{ is a Hermitian matrix}$$

$$\begin{aligned} & \left. \begin{array}{l} \text{g) } A \text{ is real matrix} \Rightarrow \bar{A} = A \\ \quad \quad \quad A \text{ is sym} \Rightarrow A^T = A \end{array} \right\} \Rightarrow \bar{A}^T = A \\ & \Rightarrow A = A^T \quad \quad \quad \Downarrow \\ & \quad \quad \quad A^H = A \end{aligned}$$

$$A(i, j)$$

$$A(i, i) \rightarrow \text{diagonal elements}$$

$$\begin{bmatrix} \textcircled{1} & 2 & 1 \\ a & \textcircled{2} & 0 \\ 0 & a & \textcircled{3} \end{bmatrix} \rightarrow U^T \xrightarrow{\text{transpose}} L^T$$

$$\begin{bmatrix} \textcircled{1} & 0 & 0 \\ 2 & \textcircled{1} & 0 \\ 1 & 1 & \textcircled{1} \end{bmatrix} \rightarrow L^T$$

$$\begin{array}{c}
 \downarrow \text{Transpose} \\
 U^T \\
 x^T \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^n \quad \underline{\underline{\mathbb{R}^{m \times n}}}
 \end{array}$$

96. A is a real square sym matrix of order n such that

$$\begin{array}{c}
 x^T A x > 0 \quad \forall \text{ non zero } x \in \mathbb{R}^n \\
 \begin{array}{ccc}
 \downarrow & \downarrow & \downarrow \\
 1 \times n & n \times n & n \times 1 \\
 \hline
 1 \times n & & 1 \times 1
 \end{array}
 \end{array}$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}^T = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 3x_1 + x_2 & x_1 + 2x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= 3x_1^2 + x_1x_2 + x_1x_2 + 2x_2^2$$

$$= \underbrace{3x_1^2 + 2x_1x_2 + 2x_2^2}_{2x_1^2 + x_1^2}$$

$$= \underbrace{2x_1^2}_{>0} + \underbrace{(x_1 + x_2)^2}_{\geq 0} + \underbrace{x_2^2}_{>0} > 0$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} x_1 + 2x_2 = 12 \rightarrow \textcircled{1} \\ 2x_1 + x_2 = 12 \rightarrow \textcircled{2} \end{cases}$$

$$\begin{cases} 3x_1 + 3x_2 = 24 & \textcircled{1} + \textcircled{2} \\ x_1 - x_2 = 0 & \textcircled{2} - \textcircled{1} \end{cases}$$

$$\begin{aligned} x_1 + x_2 &= 8 \quad \text{eq (i) + k eq (j)} \\ \underline{x_1 - x_2} &= 0 \quad x_1 = x_2 \end{aligned}$$

permissible actions

Multiplying by a non zero scalar

Interchanging the order of eqn

Adding a multiple of 1 eqn to another

$$\begin{array}{ccc} A x & \cdot & x^T \\ \downarrow & & \downarrow \\ n \times n & \times & n \\ \hline n \times 1 & & 1 \times n \end{array}$$

$$\begin{aligned} 3x_1 + 2x_2 + 3x_3 &= 1 \quad \textcircled{2} \\ 4x_1 + 6x_2 - x_3 &= 7 \quad \textcircled{1} \\ \hline 2x_1 + 8x_2 + x_3 &= 12 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 3 & 2 & 3 & 1 \\ 4 & 6 & -1 & 7 \\ 2 & 8 & 1 & 12 \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 12 \end{bmatrix}$$

$$\begin{array}{c} \text{coef matrix} \quad \text{var vector} \\ \swarrow \quad \searrow \\ A x = b \\ \swarrow \quad \downarrow \quad \searrow \\ m \times n \quad n \times 1 \quad m \times 1 \\ \downarrow \quad \downarrow \\ \text{H eqns} \quad \text{\# var.} \end{array} \quad \begin{array}{c} \text{constant} \\ \text{vector} \end{array}$$

Eqns #var.

- Interchanging of rows.
- Multiplying a row by a nonzero constant
- Adding a multiple of row to another row.

$$\begin{aligned}x_1 + 2x_2 &= 12 \\ 2x_1 + x_2 &= 12\end{aligned}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ 12 \end{bmatrix}$$

$$R_1: R_1 + R_2$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 24 \\ 12 \end{bmatrix}$$

→ compact form

$$\left[\begin{array}{cc|c} 1 & 2 & 12 \\ 2 & 1 & 12 \end{array} \right]$$

→ Augmented matrix

$$[A|b]$$

↓

$$\underline{\underline{Ax=b}}$$

Elementary row transⁿ

$$\begin{aligned}2x_1 + 2 &= 3 \Rightarrow x_1 = 1/2 \\ 2x_1 + x_2 &= 3 \leftarrow \\ \underline{\underline{x_2}} &= 2\end{aligned}$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

↓
REF

A or $[A|b] \rightarrow$ rectangular matrix
↓ row op.

REF $[A|b]$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2: R_2 + R_3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3: R_3 - R_2}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - R_2} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 + 2R_2, R_2 + R_3} \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \checkmark$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \times$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \end{bmatrix}$$

$$\begin{aligned} x_1 + 2x_2 &= 3 \\ 4x_1 + 8x_2 &= 12 \\ \hline x_1 + 2x_2 &= 3 \end{aligned}$$

$$R_2 - 4R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$R_2/2, R_3 \times -1$$

p. ele $\leq$ # rows^m
p. ele $\leq$ # colⁿ

$\# \text{ p. ele} \leq \# \text{ rows}$
 $\# \text{ p. ele} \leq \# \text{ col}$
 $\# \text{ p. ele} \leq \min(m, n)$
 $\# \text{ non zero rows in REF}$
 $\downarrow$
 rank of matrix.

$A \xrightarrow{\text{row ops}} \text{REF}(A)$
 $\downarrow$
 $\text{Count } \# \text{ p. ele.}$
 $\text{rank}(A)$

$m \text{ eqns } \& n \text{ unknowns}$
 $\downarrow$
 $Ax = b$
 $\swarrow \quad \downarrow \quad \searrow$
 $m \times n \quad n \times 1 \quad m \times 1$

$\downarrow$
 $\text{no soln} \quad \text{soln exists}$

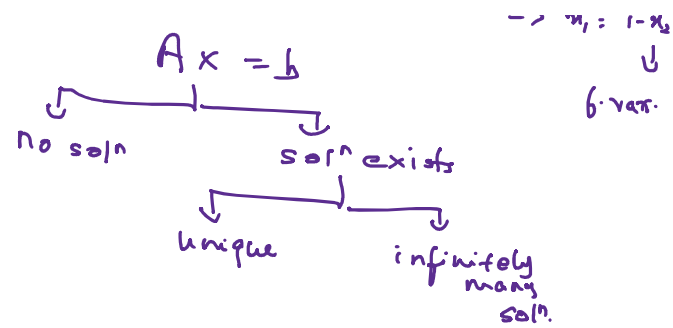
e.g:
 $x_1 + x_2 = 1 \rightarrow \textcircled{1}$
 $2x_1 + 2x_2 = 3$
 $\textcircled{1} \times 2$
 $2x_1 + 2x_2 = 2$

$\left. \begin{array}{l} 2x_1 + 2x_2 = 3 \\ 2x_1 + 2x_2 = 2 \end{array} \right\}$

unique soln
 $x_1 + 2x_2 = 12$
 $2x_1 + x_2 = 12$
 $x_1 = x_2 = 4$

inf. m. soln
 $x_1 + x_2 = 1$
 $2x_1 + 2x_2 = 2$
 $\hookrightarrow$
 $x_1 + x_2 = 1$
 $\Rightarrow x_1 = 1 - x_2$
 $\downarrow$
 h. var.

$Ax = b$



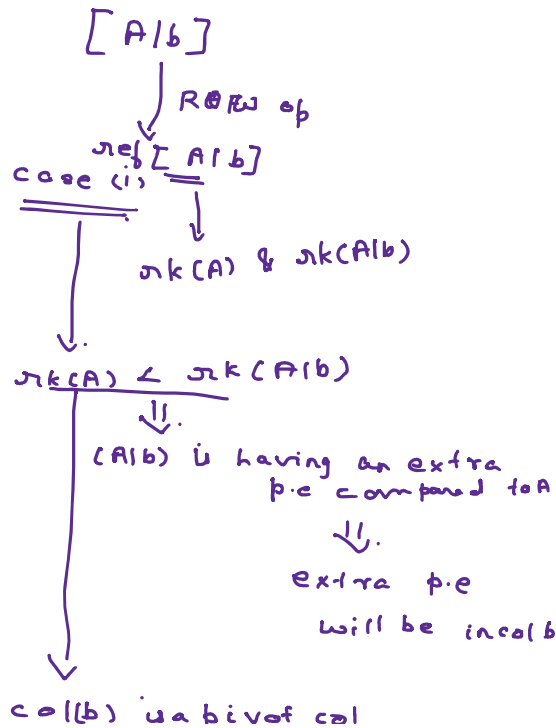
can $AX = b$ have exactly
2 distinct soln.?

$$Ax_1 = b$$

$$Ax_2 = b$$

$$x_1 \neq x_2$$

H.W.



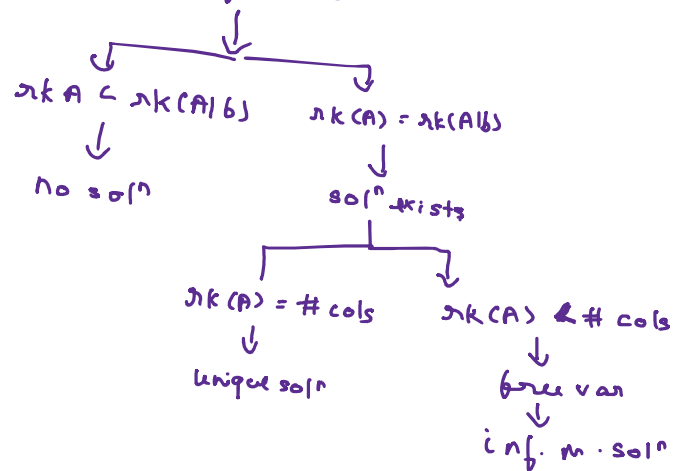
$\downarrow$ write as $(A|b)$
 $col(b)$ is a pivot col

$$\rightarrow \left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & \neq 0 \end{array} \right]$$

$$\Downarrow \\ 0 \cdot x_1 + \dots + 0 \cdot x_n \neq 0$$

$\rightarrow \leftarrow$
 $\Downarrow$
 sys is inconsistent

$$[A|b] \xrightarrow{\text{row op}} \text{ref}[A|b]$$



$$A = [a]_{1 \times 1} \quad \det A = a$$

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det B = ad - bc$$

$$\text{Minor } M_{11} = \det[d] = d$$

$$M_{12} = \det[c] = c$$

$$M_{21} = b$$

cofactor

$$C_{11} = (-1)^{1+1} \cdot M_{11} = d$$

$$C_{12} = (-1)^{1+2} \cdot M_{12} = -c$$

$$C_{21} = -b$$

$$C_{22} = a$$

$$M_{12} = \det[c] = c$$

$$M_{21} = b$$

$$M_{22} = a$$

$$C_{12} = -c$$

$$C_{21} = -b$$

$$C_{22} = a$$

$$\det = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$$

$$\det B = ad + b(-c)$$

$$= ad - bc$$

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$\det C = 1 \cdot C_{11} + 2 \cdot C_{12} + 3 \cdot C_{13}$$

$$M_{11} = \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} = 2 - 0 = 2$$

$$C_{11} = 2$$

$$\det A$$

$$= \sum_{i=1}^n a_{ii}C_{ii}$$

$$M_{12} = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

$$M_{ij}$$

$$C_{12} = (-1)^{1+2} C_{12} = 1$$

$$M_{13} = \begin{vmatrix} 0 & 2 \\ 1 & 0 \end{vmatrix} = -2$$

$$\det = 2 + 2 + 3(-2)$$

$$= -2$$

$$C_{13} = -2$$

$$M_{ij}$$



$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$\left. \begin{aligned} \det A &= \underline{a_{i1}} \underline{C_{i1}} + \dots + a_{in} C_{in} \\ &= a_{ij} C_{ij} + \dots + a_{nj} C_{nj} \end{aligned} \right\}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 6 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \det = 0$$

$$\det \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$= 1 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

$$+ 1 \begin{vmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

————— X ————— X —————

$$v = c_1 v_1 + \dots + c_m v_m$$

↓

$c_i \in \mathbb{R}$ or scalars

L.C of
 $v_1, \dots, v_m$

$v_i \in \mathbb{R}^n$
 $\forall i = 1 \dots m$

$$\Rightarrow v = \begin{bmatrix} v_1 & \dots & v_m \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

↓
 $n \times 1$

↓
 $n \times m$

↓
 $m \times 1$

clearly $c_i = 0 \quad \forall i = 1 \dots m$

$$\Rightarrow v = 0$$

$\Rightarrow 0 \in \mathbb{R}^n$ vector can always
be expressed as L.C
of any given
vectors in $\mathbb{R}^n$

Trivial L.C

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Here not all coeff are

$$\text{zero} \Rightarrow \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\}$$

as L.D

If I can find $c_i, i=1 \dots m$

such that not all c_i are zero

s.t. $\sum c_i v_i = 0$ then

$\{v_1, \dots, v_m\}$ is o L.D set

$$\underbrace{\begin{bmatrix} v_1 & \dots & v_m \end{bmatrix}}_A \underbrace{\begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}}_X = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$AX = 0$$

$\Downarrow$

$\exists$ at least 1 solⁿ $\Rightarrow$ This system is consistent

Q: Is the solⁿ unique

Solⁿ is unique $\rightarrow$ Set $\{v_1, \dots, v_m\}$ is L.I

Solⁿ is not unique $\rightarrow$ Set $\{v_1, \dots, v_m\}$ is L.D

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$$

is a L.D
set

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

is a L.I

$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A \cdot X = 0$$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 2 & 1 & 0 \end{array} \right]$$

$$R_2 - 2R_1$$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & -3 & 0 \end{array} \right]$$

↓

$$rk = 2 = \# \text{ var}$$

⇓

unique solⁿ

↓

$$a = 0, b = 0$$

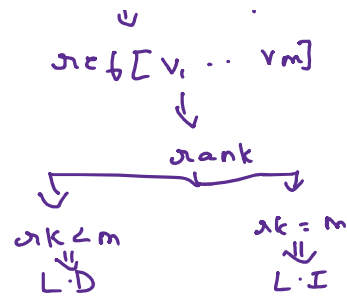
$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\} \rightarrow \text{L.I}$$

$$\{ v_1, \dots, v_m \}$$

↓

$$[v_1, \dots, v_m]$$

↓ row ops



$$c_1 v_1 + \dots + \boxed{c_j} v_j + \dots + c_m v_m = 0 v$$

Suppose $\{v_1, \dots, v_m\}$ is a L.D set

$$\Rightarrow \exists c_1 \dots c_m \in \mathbb{R} \text{ s.t. } \sum_{i=1}^m c_i v_i = 0$$

$$\text{h. } \exists j \in \{1, \dots, m\} \text{ s.t. } c_j \neq 0$$

$$\begin{aligned}
 \sum_{\substack{i=1 \\ i \neq j}}^m c_i v_i &= -c_j v_j \\
 \text{h. } -c_j &\neq 0 \\
 \sum_{\substack{i=1 \\ i \neq j}}^m \left(\frac{-c_i}{c_j}\right) v_i &= v_j
 \end{aligned}$$

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix} \right\} \rightarrow \text{L.D}$$

$$\left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \right\}$$

$$\begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{bmatrix}_{2 \times 3}$$

$$\downarrow \text{rank} \leq 2 < 3$$

⇓
L.D

In general, p vectors from $\mathbb{R}^n$

forms a L.D set
if $p > n$

Given: $\{v_1, \dots, v_m\} \rightarrow$ L.D

$[v_1 \ v_2 \ \dots \ v_m]$

↓ row ops

def $[v_1 \ \dots \ v_m]$

original vectors corr to nb
cols can always be
expressed as L.C of
vectors ~~base~~ to p cols
to the left of it

$\Omega \rightarrow$ S.S.

Toss 2 coins if the coins are fair

$\Omega = \{ \underbrace{HH}_{1/4}, \underbrace{HT}_{1/4}, \underbrace{TH}_{1/4}, \underbrace{TT}_{1/4} \}$

$X(\omega) = \#H \quad \forall \omega \in \Omega$

⇓

$X: \Omega \rightarrow \mathbb{R}$

$X(HH) = 2 \quad X(HT) = X(TH) = 1$
 $X(TT) = 0$

$X: \Omega \rightarrow \mathbb{R} \rightarrow$ S.V

$X \Rightarrow$

0	$\rightarrow P(X=0) = P(TT) = 1/4$	} Distri bution
1	$\rightarrow P(X=1) = P(HT) + P(TH) = 1/2$	
2	$\rightarrow P(X=2) = P(HH) = 1/4$	
$P(X=k) = 0 \quad \forall k \neq 0, 1, 2$		

$$\begin{aligned} &\hookrightarrow P(X=2) = P(H=1) = \frac{1}{4} \\ &P(X=k) = 0 \quad \forall k \neq 0, 1, 2 \end{aligned} \quad \Bigg\}$$

Primal prob

$$\text{Max } z = c^T x$$

s.t. c

$$Ax \leq b$$

$$x \geq 0$$

Dual prob

$$\text{Min } W = b^T y$$

s.t. c

$$A^T y \geq c$$

$$y \geq 0$$